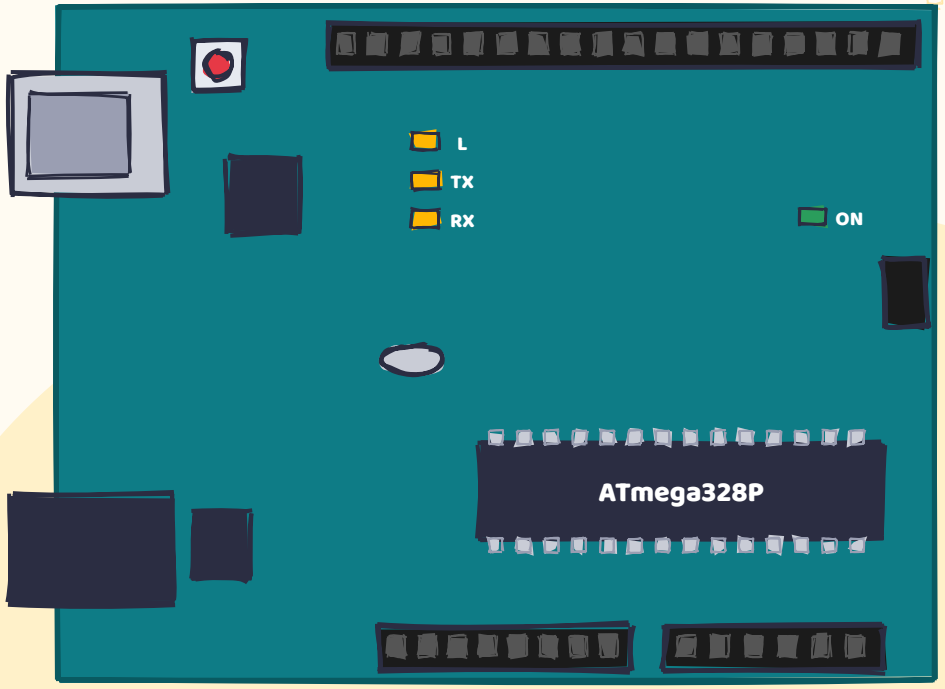
Foundation Series · Board 1

Arduino Uno Basics

Meet the tiny computer behind thousands of projects

ATmega328P | Pins, power & memory | First sketches



Did you know?
Arduino was created in 2005 at a design school in Ivrea, Italy, to help students build electronics projects cheaply.

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Look around you

Tiny computers are hiding everywhere



Washing machine

Reads the dial, runs the motor and the timer



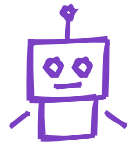
Microwave

Counts down, beeps, controls the heater



Car

A modern car has dozens of tiny computers



Toys and robots

Read buttons and sensors, drive motors

None of these has a keyboard or screen like a laptop. **So what is controlling them?** A **microcontroller**.

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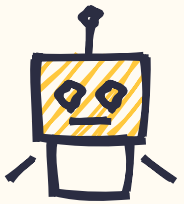
Learning aims

Today I will learn to...

1. Explain what a **microcontroller** is and what Arduino means
2. Name the main **parts of the Uno board**
3. Use **digital, PWM and analog** pins correctly
4. Power the board **safely**
5. Write, upload and understand a **first sketch**

Keywords

microcontroller · pin · digital · analog · PWM · sketch · upload · GND



LEARN

Watch and think

Video time!



Arduino Explained for Beginners in 3 Minutes

Danovation · about 4 minutes · youtu.be/aRT3kPGD9U0

Watch for…

1. What is an Arduino: a computer, a sensor or a **microcontroller**?
2. What are **input** and **output** pins used for?
3. How does the code get from the laptop to the board?

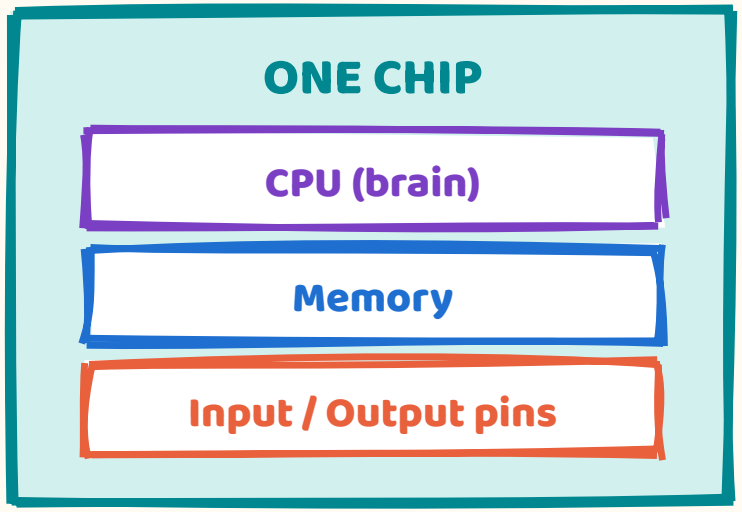
Scan to watch

After the video: Point to the USB socket, the power pins and the digital pins on your own Arduino Uno.

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The big idea

What is a microcontroller?



	Arduino Uno	Laptop
Job	Runs one program	Runs many apps
Speed	16 MHz	Billions of Hz (GHz)
Memory	2 KB SRAM	Billions of bytes (GB)
Power	About 0.25 W	Tens of watts
Talks to	LEDs, sensors, motors	Screen, keyboard, internet
Starts up	Instantly	Needs to boot an OS

A whole small computer **on one chip**: it runs **one program**, forever, from the moment it gets power.

Not just a board

What does 'Arduino' mean?



The board

Hardware like the Uno, Nano and Mega



The software

Arduino IDE, where we write and upload code



The language

C/C++ with easy commands like digitalWrite()



The community

Millions of free examples and libraries

Open source

The designs are shared freely, so many companies make **compatible boards** (clones). They work the same way.

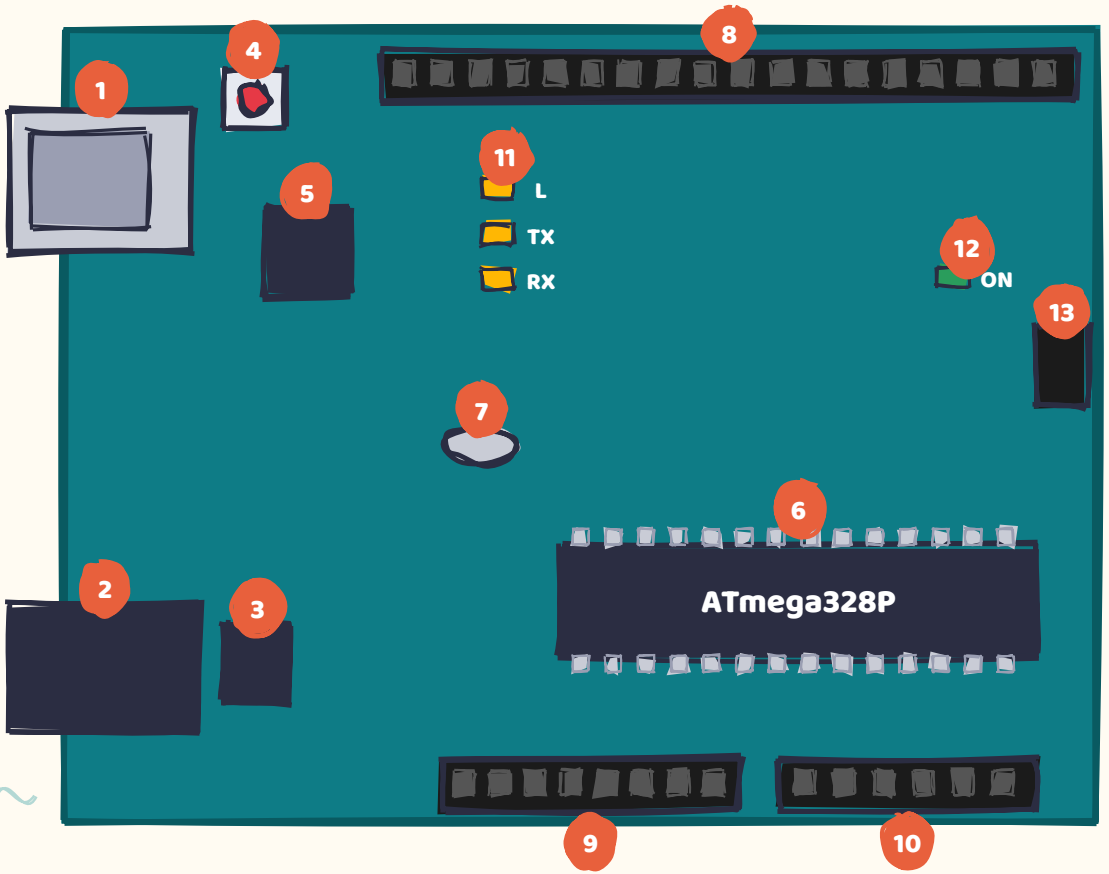
In real life

Artists, scientists and engineers use Arduino to build prototypes quickly.

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Take a tour

Parts of the Arduino Uno R3



1. **USB-B port** — Power + upload code
2. **DC barrel jack** — 7-12 V from adapter
3. **Voltage regulator** — Makes a steady 5 V
4. **Reset button** — Restarts your sketch
5. **ATmega16U2** — USB-to-serial translator
6. **ATmega328P** — The main brain
7. **16 MHz resonator** — The heartbeat clock
8. **Digital pins 0-13** — Top header
9. **Power pins** — 5V, 3.3V, GND, Vin
10. **Analog pins A0-A5** — Read sensor voltages
11. **L, TX, RX LEDs** — L is on pin 13
12. **ON LED** — Board has power
13. **ICSP header** — Direct chip programming

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Inside the brain

The ATmega328P microcontroller

Type	8-bit AVR microcontroller (Microchip)
Clock speed	16 MHz: 16 million steps every second
Flash memory	32 KB (0.5 KB used by the bootloader)
SRAM	2 KB
EEPROM	1 KB
Operating voltage	5 V
I/O current	20 mA recommended, 40 mA absolute max per pin
Total chip current	200 mA max through VCC and GND

Memory, explained

Flash = the book. Your program lives here, even with power off.

SRAM = the whiteboard. Variables while running, wiped at power off.

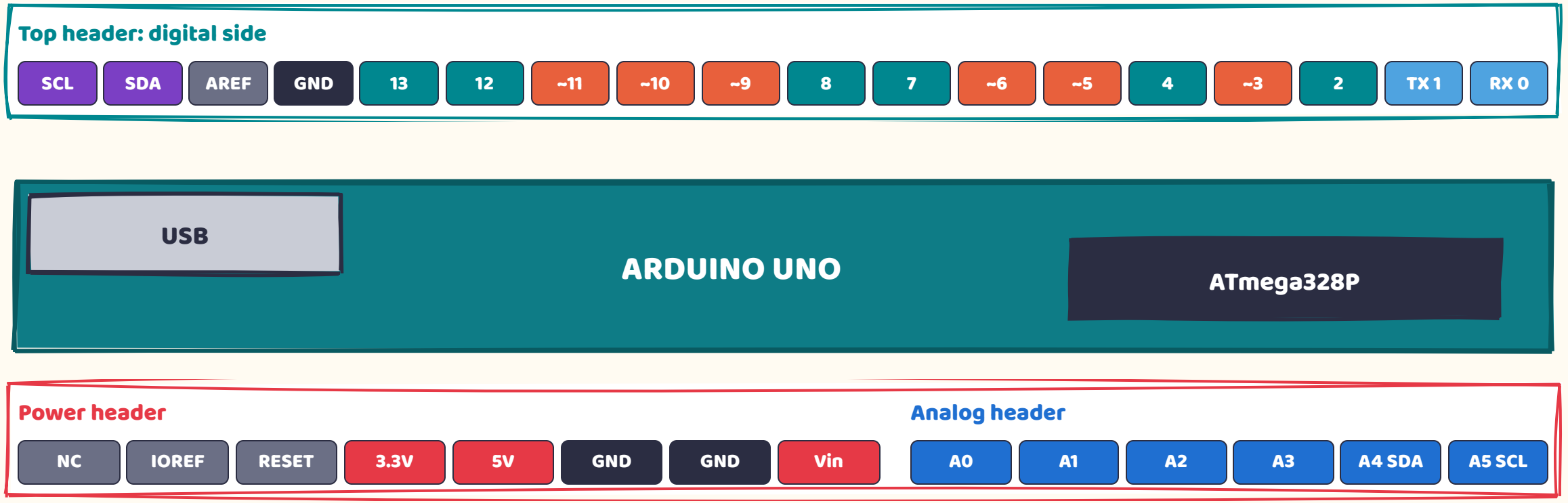
EEPROM = the notebook. Small data you want to keep, like a high score.

Why so small?

2 KB is tiny, but enough for thousands of sensor projects. Big text and images do not fit!

Know your pins

The Uno pin map



Pins 0–13

Digital pins: ON or OFF

Only two states

HIGH = 5 V (on)

LOW = 0 V (off)

Like a light switch: nothing in between.

Mode	What it does	Used for
OUTPUT	The pin **gives** 5 V or 0 V	LED, buzzer, relay
INPUT	The pin **reads** HIGH or LOW	Button, PIR, IR sensor
INPUT_PULLUP	Input with an internal pull-up resistor (about 20-50 kΩ)	Button to GND, no extra resistor

```
pinMode(13, OUTPUT);      // set up once, in setup()
digitalWrite(13, HIGH);   // LED on
int b = digitalRead(2);   // HIGH or LOW
```

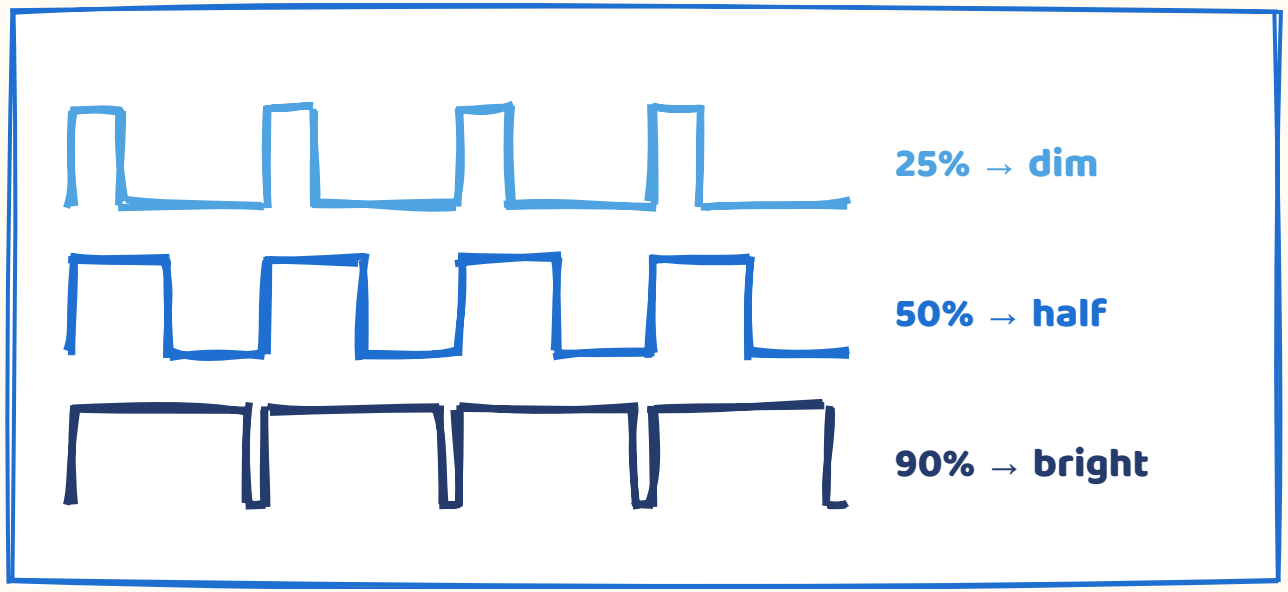
Golden rule

Every LED needs a resistor (220 Ω). Each pin: 20 mA is safe, 40 mA can destroy it.

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Pins with a ~

PWM: pretending to be analog




```
analogWrite(9, 0);     // always off
analogWrite(9, 128);   // about 50%
analogWrite(9, 255);   // always on
```

How it works

The pin switches ON and OFF very fast. The **percentage of ON time** (duty cycle) sets the average power.

PWM pins: **3, 5, 6, 9, 10, 11**

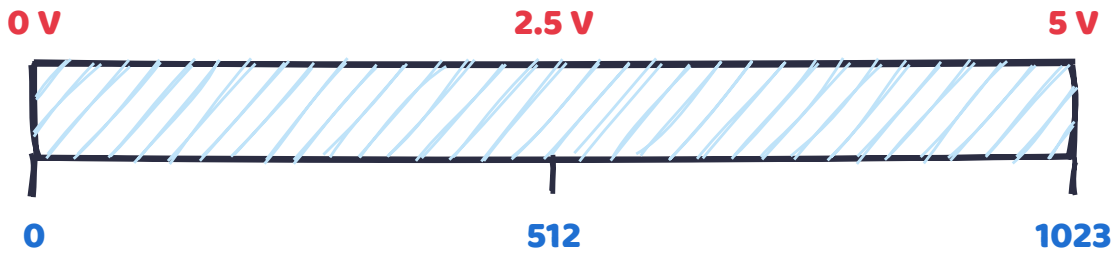
Speed facts

About 490 times a second on most PWM pins, 980 on pins 5 and 6. Too fast for our eyes to see!

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Pins A0–A5

Analog inputs: reading voltages



Resolution	10-bit ADC: values 0 to 1023
Range	0 V to 5 V (default reference)
One step	about 4.9 mV
Convert	volts = reading × 5.0 / 1023

```
int raw = analogRead(A0);    // 0-1023
float v = raw * 5.0 / 1023.0;
Serial.println(v);           // volts
```

Worked example

analogRead gives **512**

512 × 5.0 / 1023 = **2.50 V**

Perfect for LDRs, potentiometers and soil sensors.

A0-A5 can also work as extra digital pins (14-19).

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Talking to other chips

Special pin jobs

Bus	Pins	Used for
Serial (UART)	0 = RX, 1 = TX	Shared with USB. Keep free while uploading.
I2C	A4 = SDA, A5 = SCL	OLED, LCD, MPU6050: many devices, 2 wires
SPI	10 SS, 11 MOSI, 12 MISO, 13 SCK	SD cards, RFID RC522, fast displays
Interrupts	Pins 2 and 3	React instantly to a pulse or button

Serial Monitor

Your window into the board. Print numbers from sensors to the laptop screen.

```
Serial.begin(9600);
Serial.println("Hi!");
```

Common bug

Upload fails with a module on pins 0 and 1? Unplug it, upload, then plug it back.

Learn from others

Six mistakes every beginner makes

LED with no resistor

Burns the LED or the pin. Use 220 Ω.

Motor on a pin

Pins give 20-40 mA. Motors need hundreds. Use a driver.

Wrong COM port

Upload fails. Re-select Tools → Port.

Things on pins 0/1

Upload errors. Keep them free.

No common GND

Sensors give random values. Join all GNDs.

Wiring with power on

Sparks and shorts. Unplug USB first!

Build, check, then power. Most "broken" Arduinos are really a loose wire or a wrong pin.

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Meet the family

Same idea, different sizes

Arduino Nano

Same ATmega328P, tiny size, fits a breadboard

Arduino Mega 2560

54 digital pins, 16 analog, 256 KB flash: for big robots

Good news

Code you write for the Uno works on most of these with little or no change.

Uno R4 Minima

32-bit Renesas RA4M1, 48 MHz, 256 KB flash, 32 KB SRAM

Uno R4 WiFi

R4 plus an ESP32-S3 for Wi-Fi and Bluetooth, and a 12x8 LED matrix

Need Wi-Fi? The Uno R3 has none. Next board: the **ESP32**, with Wi-Fi and Bluetooth built in.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Powering the Uno
2. From code to a working board
3. Blink: the Hello World of hardware
4. A button with INPUT_PULLUP

Stay safe

Powering the Uno

USB cable	**5 V from laptop**	Protected by a 500 mA resettable fuse
Barrel jack	**7-12 V recommended**	Absolute limits 6-20 V; the regulator gets hot at high voltage
Vin pin	**7-12 V in**	Same as the jack, no reverse protection
5V pin	**Output 5 V**	Only feed it a clean regulated 5 V, never more
3.3V pin	**Output 3.3 V**	Maximum 50 mA

Motors?

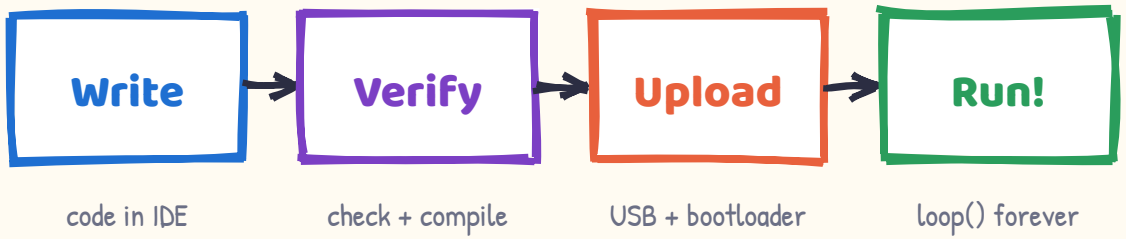
Never power motors from a pin or the 5V pin. Use a **driver** (L298N) and a **separate supply**, and connect the GNDs together.

Short circuit!
Never touch 5V to GND with a wire.

Rule of thumb: for LEDs, sensors and small modules, USB power from the laptop is enough.

Arduino IDE

From code to a working board




```
void setup() {
  // runs ONCE when the board starts
}

void loop() {
  // runs AGAIN and AGAIN, forever
}
```

Before you upload

- ☑ Tools → Board → **Arduino Uno**
- ☑ Tools → Port → the **COM** port that appears
- ☑ Click the tick to **verify**, the arrow to **upload**

Vocabulary

A program for Arduino is called a sketch. Arduino also runs in the browser with Arduino Cloud Editor.

MAKE

Your first sketch

Blink: the Hello World of hardware

```
void setup() {
  pinMode(LED_BUILTIN, OUTPUT);  // pin 13
}

void loop() {
  digitalWrite(LED_BUILTIN, HIGH); // on
  delay(1000);                     // wait 1 s
  digitalWrite(LED_BUILTIN, LOW);  // off
  delay(1000);
}
```

Line by line

LED_BUILTIN is the L LED on pin 13.

pinMode makes the pin an output.

digitalWrite sets 5 V or 0 V.

delay(1000) waits 1000 ms = 1 s.

Try it

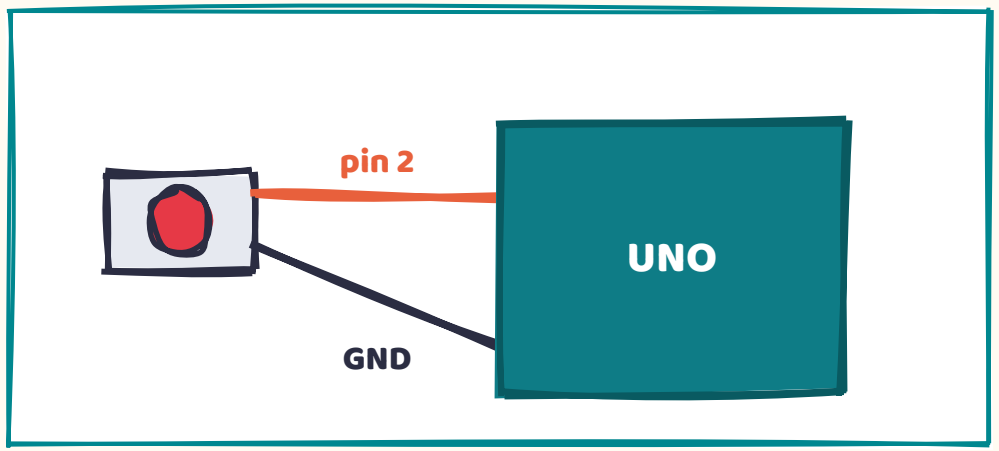
Change both delays to 100. What happens? Now make a heartbeat: on 100, off 100, on 100, off 700.

Reading input

A button with INPUT_PULLUP

```
void setup() {
  pinMode(2, INPUT_PULLUP);
  pinMode(13, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int s = digitalRead(2);  // pressed = LOW
  digitalWrite(13, s == LOW);
  Serial.println(s);
  delay(50);
}
```



Why LOW = pressed?

The pull-up holds the pin at 5 V (HIGH). Pressing the button connects it to GND, so it reads **LOW**.

Show what you know

Recap and quick quiz

Remember

- [x] ATmega328P: 16 MHz, 32 KB flash, 2 KB SRAM, 1 KB EEPROM
- [x] 14 digital pins, 6 with PWM (~), 6 analog inputs
- [x] Digital = 0 V or 5 V, analog read = 0 to 1023
- [x] setup() runs once, loop() runs forever

Quick quiz

1. Which pins can use analogWrite()?
2. What does analogRead() give for 2.5 V?
3. Why do pins 0 and 1 cause upload problems?
4. Why does an LED need a resistor?

Challenge

Blink SOS in Morse code on the L LED: 3 short, 3 long, 3 short flashes.

Next: ESP32 Basics, the Wi-Fi and Bluetooth board.

